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22AIE401 – Reinforcement Learning

Risk-Aware Adaptive API Rate Limiting Using Reinforcement Learning

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Problem Background

The current applications depend significantly on Application Programming Interfaces (APIs) for communicating among clients, services, databases, clouds, and micro services. In cases where an API receives many requests at once from different clients, traffic control becomes one of the requirements for the sustainability of the application.

The traditional methods of controlling the rate of APIs generally apply pre-set thresholds like limiting the number of requests for a client in a certain time interval. Even if Token Bucket, Leaky Bucket, and Sliding Window techniques manage the traffic effectively, they tend to work using predefined or manually set limits.

The problem is that the actual traffic is dynamic, where an increase in requests might mean either legitimate bursts in traffic like simultaneous requests by many clients or an attack. The same limit applied in both cases could restrict legitimate users or even fail in protecting the back-end system.

Moreover, the right rate limit depends on the state of the backend. A state of high CPU usage, higher latency and usage of memory and increasing queue of requests may show that there is a need for reducing the amount of traffic to the server.

Reinforcement Learning is an approach in which decisions are learned through the interactions with the environment. The research project is focused on applying reinforcement learning to adaptive API rate limiting when the agent learns to adapt traffic control policy based on the client behavior and system state**.**

Problem Statement

Most API gateways employ rate limiting methods based on pre-defined rate limits that allow clients to generate a certain number of requests. Even though these techniques are quite effective, rate limiting that doesn't change in response to various conditions is hard to apply to variable traffic patterns, different client actions, or backend system conditions.

For instance, if there is a legitimate client who suddenly generates a burst of requests, the client will be regarded the same way as the one who generates malicious traffic. Moreover, the traffic rate limit won't change even if the backend system suffers from increased CPU utilization or latency.

Thus, it is crucial to have an adaptive traffic management system that would be able to adaptively define the optimal traffic control policies depending on the current condition of the client and backend system.

As a part of the proposed project, an API gateway based on reinforcement learning will observe different parameters like request rate, traffic burstiness, client actions, server load, response latency, and error rate, and take appropriate traffic-management actions.

The system will attempt to balance:

- legitimate request access
- prevention of API abuse
- backend stability
- response latency
- efficient resource utilization
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Why Reinforcement Learning?

The problem of API traffic management itself could be easily defined as a sequential decision making problem.

At each point in time, the system is aware of the current state of API traffic and the backend environment. On the basis of this state, a decision needs to be made concerning the amount of traffic that can be allowed through.

For example :

Current State —> High traffic —> High CPU —> RL Agent —> Reduce rate limit —> Traffic decreases —> CPU and latency improve —> New State

Unlike traditional rule-based systems, where thresholds are manually defined, an RL agent can learn a policy through interaction with an API traffic environment.

The problem can be represented using:

State:
Current request rate, client risk, CPU utilization, memory utilization, latency, error rate, and current rate limit.

Action:
Increase, maintain, decrease, throttle, or block traffic.

Reward:
Reward the system for serving legitimate requests while penalizing backend overload, abusive requests, excessive latency, and unnecessary throttling.

Literature Review

S. No	Paper	Year
1	**Adaptive Rate Limiting Using Reinforcement Learning to Thwart API Abuse**	2023
2	**TopFull: An Adaptive Top-Down Overload Control for SLO-Oriented Microservices**	2024
3	**Multi-Objective Adaptive Rate Limiting in Microservices Using Deep Reinforcement Learning**	2025
4	**DAGOR: Overload Control for Scalable Microservices**	2018
5	**Breakwater: Overload Control for μs-scale RPCs**	2020
6	**Machine Learning Enhanced API Rate Limiting and Throttling Strategies in High Volume Fintech Systems**	2022

Existing System

Current API management systems commonly use predefined rate-limiting mechanisms such as Token Bucket, Leaky Bucket, Fixed Window, and Sliding Window algorithms.

Cloud platforms and API security platforms have also introduced adaptive mechanisms for detecting abnormal traffic and managing API abuse.

For example, **Cloudflare API Shield** provides adaptive mechanisms for identifying volumetric API abuse based on observed traffic behavior.

Systems such as **Envoy** also provide admission-control mechanisms that can dynamically reject requests based on observed service conditions.

Research systems such as **TopFull** investigate reinforcement-learning-based admission control for microservices and dynamically regulate API traffic according to system conditions.

However, existing systems generally address specific aspects of the problem, such as:

- fixed traffic thresholds
- anomaly detection
- backend overload
- API abuse prevention
- service-level objectives

The proposed project aims to combine **client behavioral information and backend health information** within an RL-based traffic-control framework.

Gap Analysis

The traditional methods for rate limiting APIs are based mainly on preconfigured thresholds; therefore, these methods are rather inflexible in coping with fast changes in the traffic flow. The threshold is optimal in the case of normal workload but may become too strict or too liberal when there is a change in the traffic flow.

Previously, researchers have tried to design an adaptive rate limiter based on reinforcement learning, but there are other algorithms that are designed for detecting API abuses or for microservices overload control. These solutions are usually centered on one particular aspect of the problem. Namely, API abuses detectors are aimed at finding suspicious clients.

Another significant issue concerns separating legitimate traffic from malicious traffic. Having a big number of requests from a client does not automatically imply that the client acts maliciously. Likewise, a traffic increase might be due to a genuine activity rather than being an attack. Thus, request volume may not be enough for making the proper rate limiting decision.

There is also a necessity to take into account the current state of the backend in making the decision about traffic control. The same client traffic will have different impacts based on CPU utilization, memory utilization, response time and other factors. Therefore, there is a need for having an adaptive traffic control technique that takes into account client behavior and the state of the backend.

Proposed System

The proposed project aims to develop a **Risk-Aware Adaptive API Rate Limiting system using Reinforcement Learning**. Unlike conventional API gateways that rely on fixed rate limits, the proposed system will continuously observe incoming API traffic, client behavior, and backend system conditions to determine an appropriate traffic-control policy.

The system will collect information such as request frequency, traffic burstiness, historical client behavior, error rate, suspicious request patterns, CPU utilization, memory utilization, response latency, and the current rate limit. Client behavioral information can be used to estimate a risk score, while backend monitoring provides information about the current health of the service.

These observations will be provided to an RL agent, which will learn to select an appropriate traffic-control action. The possible actions may include increasing the rate limit, maintaining the current limit, decreasing the rate limit, throttling the client, or temporarily blocking the client. The agent will receive rewards for successfully serving legitimate traffic and maintaining backend stability, while penalties will be associated with abusive requests, excessive latency, backend overload, and unnecessary throttling of legitimate users.

The proposed system will initially investigate a Q-learning-based approach and evaluate its performance using simulated API workloads. The simulation will include normal traffic, sudden traffic bursts, high-volume legitimate traffic, abusive clients, and backend overload conditions. The proposed approach will then be compared with conventional rate-limiting techniques to study its effectiveness in dynamically managing API traffic.

The overall research direction is therefore to investigate whether combining **client behavioral risk with real-time backend health in an RL-based decision-making framework** can provide more adaptive API traffic management while maintaining service stability and legitimate-user access.